

Triangular Waveform Generation using Mixed Signal Modeling

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Abstract

The development of modeling languages such as Verilog-HDL and Verilog-AMS allows the behavior of analogue and mixed signal circuits to be described with a more lucid way as compared to the conventional circuit-level simulators. Mixed-signal simulators are thus providing a new platform for more efficient modeling of mixed signal that contains both the analogue and discrete features which was not previously possible. The non-linear characteristics of many circuits are still difficult to be exactly described by the conventional simulators. Hence it is required to explore the simulation behavior of analogue and mixed-signal circuits in a more efficient manner which can be done using mixed signal modeling languages. This paper presents an alternative approach to model the behavior of analogue and mixed-signal circuits based upon piecewise linear waveforms. A triangular waveform generator is modeled using the mixed signal modeling. The proposed model of triangular waveform generator can be used in the circuits for signal processing. The triangular-wave generator is often necessary in laboratories for efficient generation of other types of fixed periodic functions. It is also useful for performance testing of electrical network and servo-systems. The waveform generator modeled using mixed-signal modeling provides more accurate and highly precise waveform which reflects its advantages.

Keywords: *Triangular waveform generators, Mixed-signal modeling, piecewise linear waveforms*

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INTRODUCTION

A triangular waveform is a non-sinusoidal waveform which resembles to triangular shape. It is a periodic, piecewise linear, continuous real function. An accurate symmetrical triangular waveform is desirable in waveform conversions such as triangular to sine and triangular to square wave conversions, which are used to study the transfer characteristics of circuits and systems. Triangular waveform generation is also useful for spread spectrum clock generation [1] or frequency modulated ultra-wide band (FM-UWB) systems.

Such application specific requirements depend on the accuracy of the magnitude and the symmetry of the triangular waveform. These requirements of triangular waveform are not particularly rigid in some of the applications but when we are concern with any particular

application specific system such as convertors like single sweep triangle-to-sine waveform converter, and then the distortion in the sinusoidal voltage output waveform generated will be directly affected by the input triangular waveform. If the input waveform is not up to the mark in the terms of amplitude and symmetry then its adverse effect will be reflected in the sine wave generated and we will be unable to achieve our intended goal. Thus, it reveals that we require the generated waveform to be highly accurate and precise in the terms of its parameters.

Schmitt trigger circuit with a linear integrator which is also referred to as free-running triangular waveform generator is a popular method of generation of the triangular waveform. This is a simple circuit, but it is not responsive to linear variation of frequency over a wide range. There exists one more

approach in which a FM modulation circuit is improved to charge and discharge a capacitor linearly through two constant current sources [2] and thus generates triangular waveform. In conventional generation method we can also use a complementary pair of transistors, connected in parallel, to alternately charge and discharge a capacitor linearly. In another approach a direct coupled, a stable multivibrator [3] can be used. All the above methods rely on charging and discharging of capacitor. Multiphase triangular waveform generation can also be done by using relaxation oscillator [4] but the propagation delay of the comparator makes it difficult to achieve precise triangular waveform.

The above mentioned methods are simple, but the exactness in the symmetry of the triangular waveform generated by these methods depends on the charging and discharging currents input to the capacitor. Hence we are unable to get lofty amount of accuracy in the generated waveform. Hence, we need a better way to express such types of analog signals so that it can fulfill our requirements.

This paper presents modeling of triangular waveform generator that can preserve the symmetry, linearity and constant amplitude of the waveform over a wide range of frequencies. The modeling of triangular waveform has been done with the help of Verilog-AMS which is intended to support modeling of analog and mixed-signal systems. The modeling produces triangular waveforms with high accuracy. It also possesses the features of smooth frequency variation with higher stability over wide range. The generated waveform parameters can be easily manipulated according to our requirements for particular application. This paper makes the following contributions.

- 1) This work models triangular waveform generator that generates a triangular waveform with perfect symmetry over a wide range of frequency while maintaining amplitude constancy.
- 2) The modeled triangular waveform is then utilized in triangular to sine and triangular to square wave conversions.

The rest of the paper is organized as follows. A brief introduction to Verilog-AMS is given in

the next section, followed by simulation results and discussion. Finally, the concluding remarks are provided in the last section.

MODELING WITH VERILOG-AMS

Verilog-A is a language for defining behavior of analog components with a high level of abstractions whereas Verilog-HDL is intended to define digital systems. Verilog-A and Verilog-HDL are basically subsets of Verilog-AMS as shown in Figure 1. Verilog-AMS inherits the ability to model both analog and digital systems in a single module. Verilog-A is a compiled language which means that its code is compiled to a binary executable program in a similar manner as it is done in implementation of built-in device models. This makes Verilog-A models to respond quickly.

Verilog-AMS HDL extends the features of the digital modeling language (IEEE STD 1364-2005 Verilog HDL) to provide a single consolidate language with both analog and digital system behavior modeling and having compatibility with the present scenarios [5]. Verilog-AMS is primarily designed to support simulation of mixed signal systems.

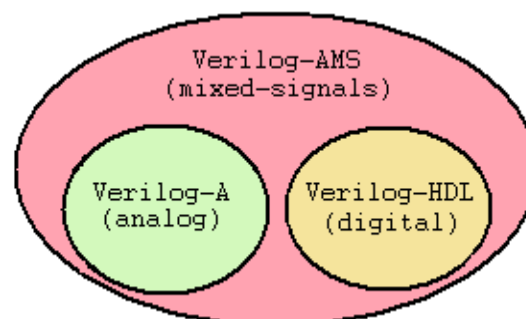


Fig. 1: Relationship between Verilog-AMS, Verilog-A and Verilog-HDL.

Below is a list of salient features of the resulting language:

- Signals of both analog and digital types can be declared in a single module in which *Initial*, *always*, and *analog* procedural blocks can appear.
- Both analog and digital signal values can be accessed i.e. it provides read capability for both analogue and digital signal in a single module.
- Analog write operations can only be done from inside an analog procedural block while digital signal values can be altered

from any context outside the analog procedural block.

- The semantics are quite similar to Verilog and discipline declaration is extended to mixed-signals.

The term 'mixed-signal' indicate systems made up of segments that process digital as well as analog signals. Verilog-A allows the description of continuous time signals and Verilog-HDL allows the description of discrete event-driven signals with abruptly changing values as shown in Figure 2. Verilog-AMS combines the features of both languages and inherits the ability to process both digital and

continuous-time analog signals in a single simulator. Thus, this language provides a single design flow that supports analog, digital and mixed-signals. This provides a better approach for the designer to deal with three different aspects in a single platform. Verilog-AMS possesses strong event-driven capabilities for analog simulation which allows signals to be expressed with ease as that of digital simulation. Thus, Verilog-AMS provides a better approach to model mixed-signal components, to create test bench and support the top down design approach with accelerated simulation speed.

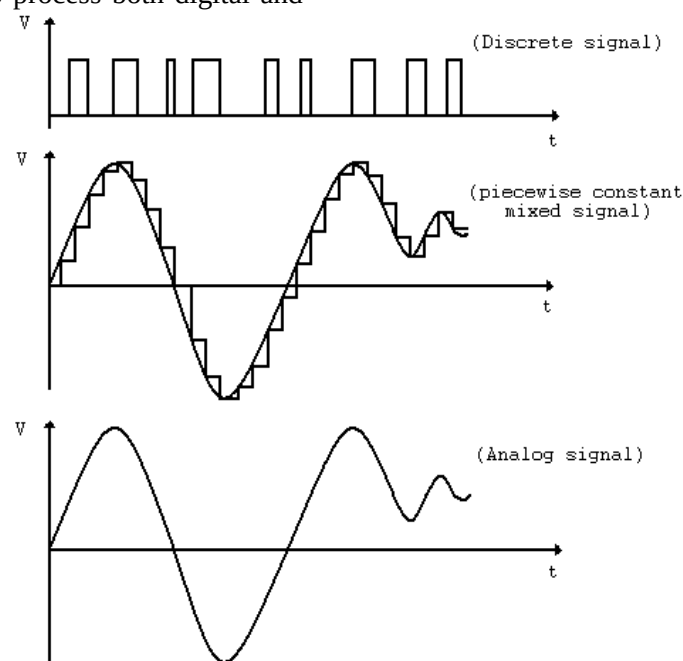


Fig. 2: Digital, Analog Discrete-event (piecewise linear) and Analog Continuous Signals.

SIMULATION RESULTS AND DISCUSSION

The focus of this work is to model triangular waveform which is further utilized in other types of wave form generations. This section presents simulation results of design metrics of triangular waveform generator which are obtained during modeling. The waveform obtained with different design aspects shows that this modeling proves to be an efficient solution for triangular waveform generation.

Triangular Waveform Modeling

A triangular waveform can be generated by utilizing anyone of the conventional method described in the section I. Triangular

waveform can also be generated in a simple manner by incorporating a digital to analog converter as shown in Figure 3. It represents the block diagram of a triangular waveform generator in which the output of a digital up-down counter, modeled using Verilog, is fed to a digital to analog converter. The triangular waveform generated using this kind of digital up-down counter exhibits higher stability and accuracy in terms of amplitude and frequency compared to that generated using analogue techniques. The internal circuit of modeled digital up-down counter, utilized here is shown in Figure 4. It consists of D flip-flops (DFF), inverters (INV), output buffers (OBUF), look-up tables (LUTs).

The LUT-4, LUT-3 and LUT-2 are further shown in Figures 4(a), (b) and (c), respectively, to describe the internal gate level implementation of the different blocks involved in the design of the waveform generator. As described above, the triangular waveform generated by the digital technique provides greater level of accuracy and precision than analogue generators. Employing this concept we model a triangular waveform generator in this paper which is done using mixed signal modeling.

The generated triangular waveform represents perfect symmetry in the waveform over a wide range of frequency while maintaining amplitude constancy. From the output

waveform it can be seen (Figure 5) that triangular waveform modeled is more precise and accurate than the triangular waveforms generated by conventional methods. The conventional methods which are generally used to generate the triangular waveform, consists of capacitor which doesn't give the accurate results as desired in certain specific applications, due to the exponential charging and discharging of the capacitor incorporated within the circuit. Modeling with Verilog-AMS mitigates these effects. Hence this modeling proves to be more linear and accurate with constant amplitude. It also proves its significance for the circuits which require input triangular waveform to be highly precise and accurate.

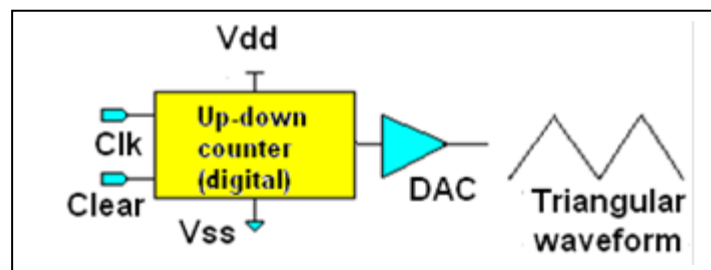


Fig. 3: A System Level Diagram of the Triangular Waveform Generator.

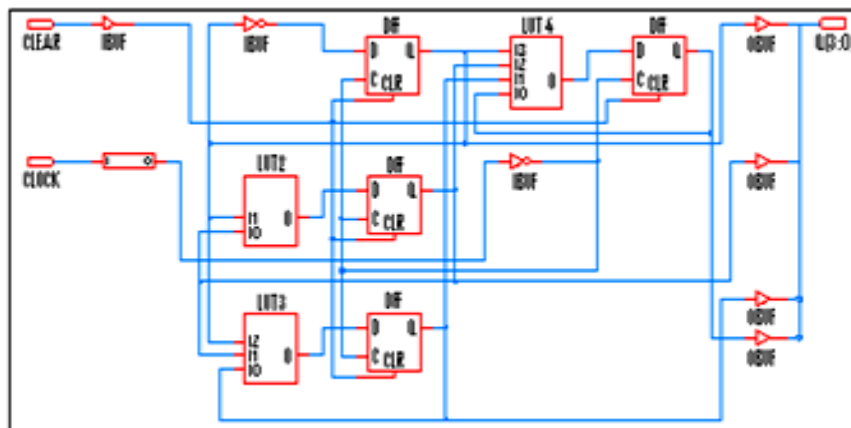


Fig. 4: Functional Block Diagram of Digital up-down Counter.

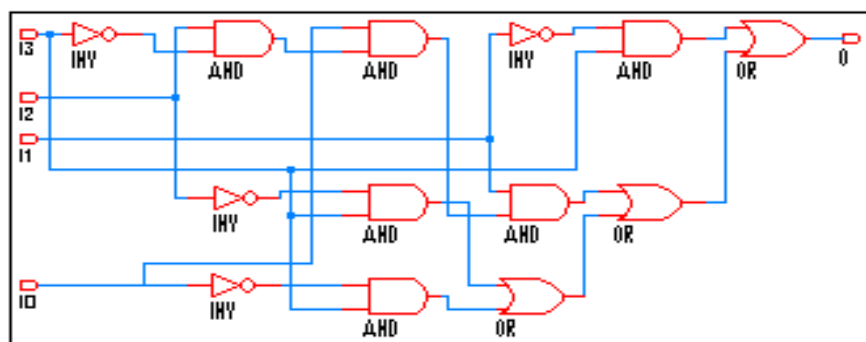


Fig. 4(a): Gate Level Implementation of LUT - 4 Shown in Figure 4.

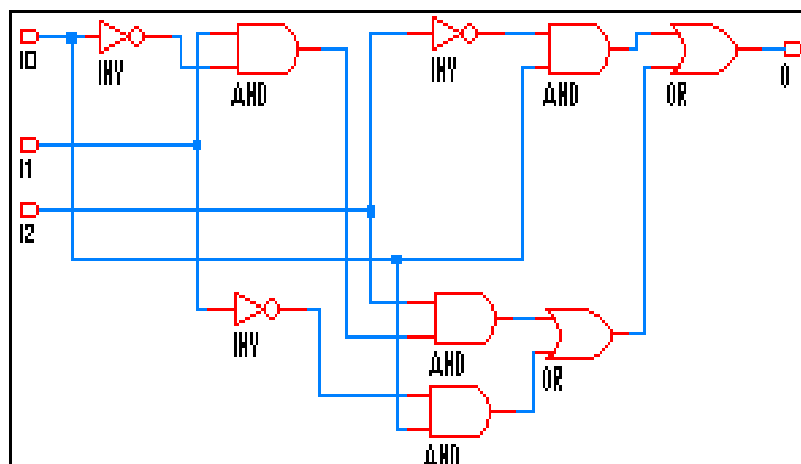


Fig. 4(b): Gate Level Implementation of LUT - 3 shown in Figure 4.

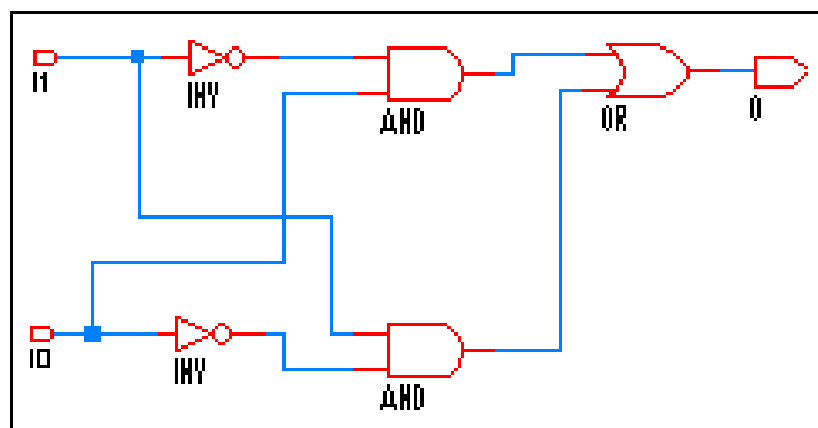


Fig. 4(c): Gate Level Implementation of LUT - 2 Shown in Figure 4.

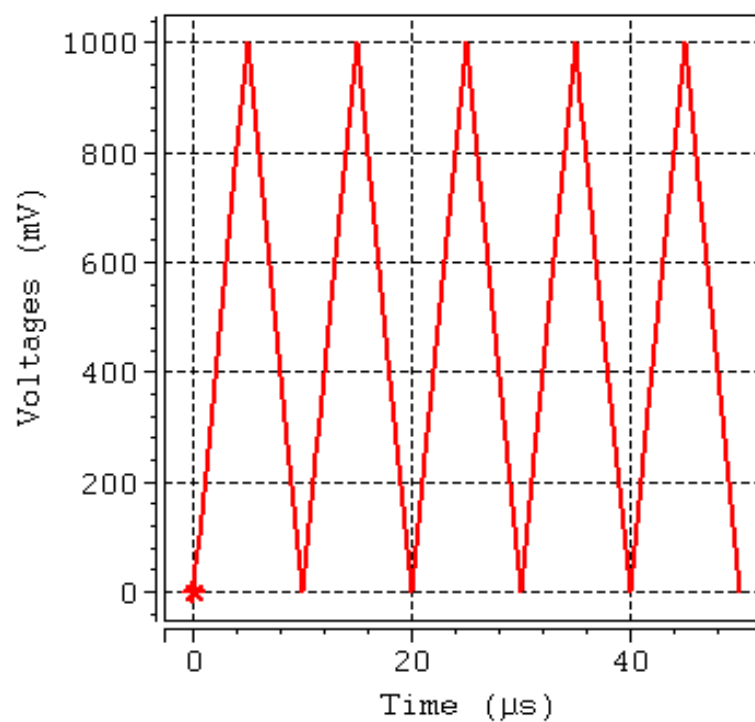


Fig. 5: Generated Triangular Waveform starting from Origin.

This modeling also provides the flexibility of variation of the parameters like amplitude, time period, rise time and fall time etc. of the waveform generated. The amplitude of the output waveform which is presented in the Figure 5 is 1000 mV and period is 10 μ s. The generated triangular waveform can also be shifted in vertical position, as per the requirements, which is shown in the output waveform exhibited in Figure 6. The waveform in Figure 6 has its voltage swing from -500 to +500 mV and having a time period of 10 μ s. This shows the ease of controllability of the parameters of the generated waveform. The waveform is

piecewise linear and has wide range of operating frequencies.

Triangular to Sine and Square Waveform Conversions

The triangular waveform has wide applications in areas such as electronic instruments, computation of fast Fourier transform for signal processing [6], telemetry, frequency modulation, input signal to different testing device, and smoothing oscillator in function generators. The usability of the circuit has been demonstrated by applying the generated waveform to a triangular to square waveform converter and triangular to sine waveform converter with addition of few components.

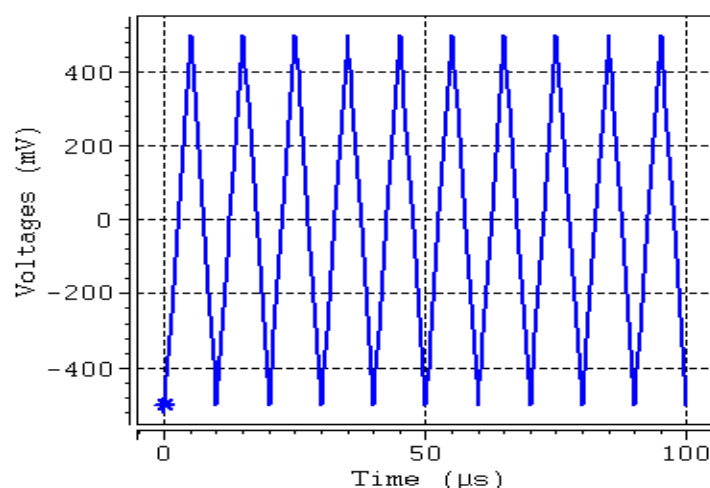


Fig. 6: Generated Triangular Waveform Symmetric about X-axis.

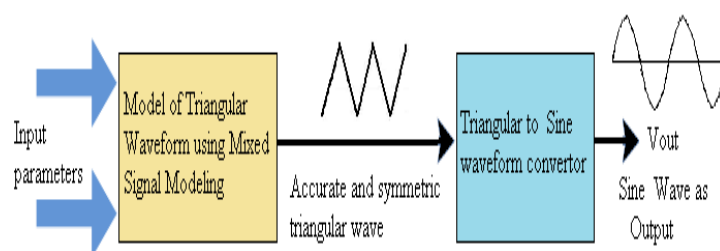


Fig. 7: Block Diagram Representation of Triangular to Sine Wave Generation using Modeled Mixed-Signal Waveform.

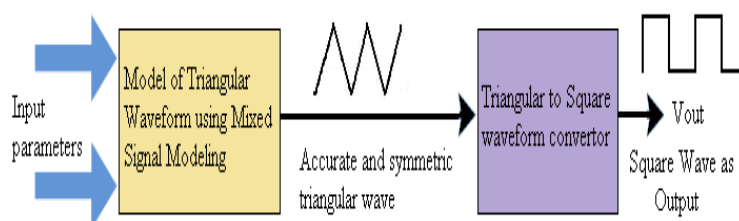


Fig. 8: Block Diagram Representation of Triangular to Square Wave Generation using Modeled Mixed-Signal Waveform.

The Figures 7 and 8 represent the methodology by which we can utilize this modeled waveform. It is obvious that to design a triangular to sine wave converter with the conventional methods, we have to incorporate some form of passive or active elements which will generate triangular waveforms. The triangular waveforms produced by these methods are applied to triangular to sine wave converter which gives sine waveform. In such cases resistors with diodes and capacitors are arranged to approximate a sine waveform with discrete time driven instances form the analog behavior. These types of signal generator are not very accurate due to their analog behavior. To increase the accuracy of the circuit we require more linear segments of highly time driven signals, which will require more number of components to be incorporated in the conventional circuits. It will also create complication in matching and synchronization of different components used. Same phenomenon is also applicable to the square waveform generation which will lack in the exactness of the desired waveform. Thus, generation and control of such waveform is of great concern for the circuit designers' point of view.

Seeing the application of these waveforms in fields of testing and calibration of electronic equipment, it is required to define reference waveform in more accurate way; otherwise the

imprecise triangular waveform will reflect its adverse effects on the generated waveform.

To remove such type of problems we have an alternative approach of modeling these signals using analog mixed signal modeling. Triangular waveform modeled in this article validates its significance as a reference input to such types of symmetric waveforms generation.

Sine wave pulse width modulator which is widely used in three phase motor drives, relies on the precision of the triangular waveform [8] given as input to it. On the basis of preciseness of the triangular waveform, the switching losses in the modulators can be forecasted. In such cases the above explained model will be very useful to drive such modulators by synchronization of triangular waveform with the sine wave. Figure 8 shows the generated output sine waveform which has been generated by utilizing the modeled triangular waveform, taken as a reference. The triangular to sine wave converter rely on the accuracy of the triangular waveform given as input to the converter. The validity and uniformity of the sine waveform can be verified from the output waveform shown in the Figure 9. We have also validated with the simulation results, the utility of the modeled triangular waveform by utilizing it as reference input to the triangular to square wave converter.

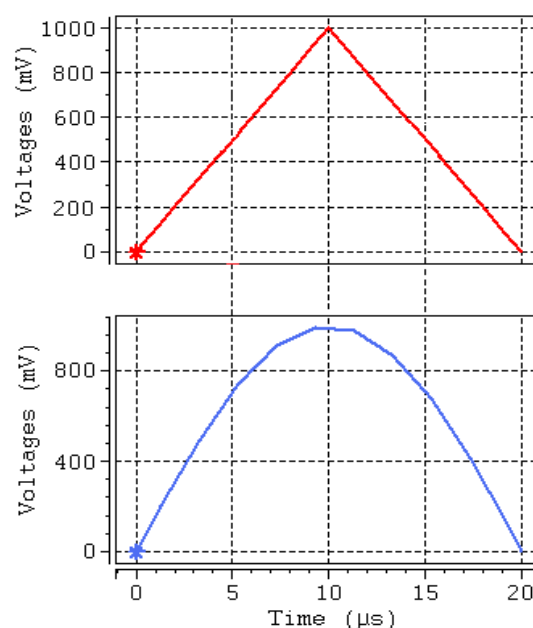


Fig. 9: Triangular to Sine Wave Converter Output Waveform.

The input modeled triangular waveform and the output square waveforms have been presented in Figure 10. The figure also exhibits that the output waveform will be highly symmetrical and accurate as it utilizes the triangular waveform modeled by mixed-signal modeling.

The paper models the triangular to sine wave convertor and triangular to square wave convertors which are performing analogue to analogue conversion modelled using Verilog-AMS. The modeling language describes the analogue behaviour of such converters in the digital domain by considering piecewise linearity in the signal. Thus analogue to analogue converter (AAC) has been modelled by capturing the analogue behaviour in digital domain using Verilog-AMS.

The variations in the output waveform with the input triangular waveform are very much synchronized. The dependency in the input waveform and the output waveform can be concluded from the overlapping input and output waveform shown in the Figure 11. The waveforms are well accurate and symmetric. As the modeling of triangular waveform has been done with the Verilog-AMS, the generated waveform provides perfect symmetry. Also the modeling of waveform is not prone to external process variation; hence, the peak to peak voltage of the generated

triangular waveform is constant with respect to the time. The waveform generated is neither decaying nor affected by harmonic distortion as can be observed from Figures 6 and 11. Therefore, the waveform is symmetric over a wider range of frequency while maintain amplitude constancy. The output frequency and symmetry of the waveform produced can be easily verified from the waveform obtained in Figure 11. Such types of waveform modeled can be utilized as a reference to many other electronic testing and measurement instruments. The precision in the output can be easily achieved as per our requirements. Moreover this modeled triangular waveform will also provide advantage of input parameters to be controlled over a wider range of order of nanoscale. This modeling produces the desirable triangular waveform in simple manner without the complexity in controlling its parameters. Generation of highly accurate and synthesized sine and square waves will help in testing and designing electronic circuits and equipments.

In this modeling we have further added few parameters like shifting of waveform along the vertical axis (see Figures 5 and 6), rise time and fall time etc. which will provide more steering power to the designer. The results of this modeling consistently show the perfectness of the behavior modeled.

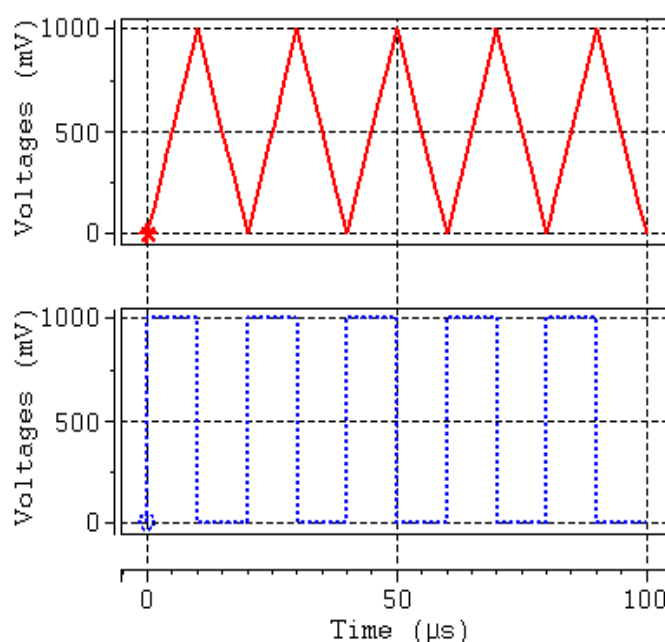


Fig. 10: Triangular to Square Wave Converter.

Triangular waveform generators with control over the output parameters such as amplitude, period, rise and fall time etc. within a wide range of operating frequency, have very vast field of applications in instrumentation and measurement systems. Such generators can be easily realized by using an operational transconductance amplifier (OTA) as a switching current source to generate charging and discharging current for the capacitor followed by a Schmitt trigger. But in these

conventional circuits the output waveforms are not that much accurate and precise as sometimes required in some specific application. An extensible approach has been described in this paper that can map the continuous signals into piecewise linear waveforms. Verilog-AMS modeling provides a foundation to model analog signals with discrete signals by generating snaps of analog sub blocks which easily fit to the discrete system specification [7].

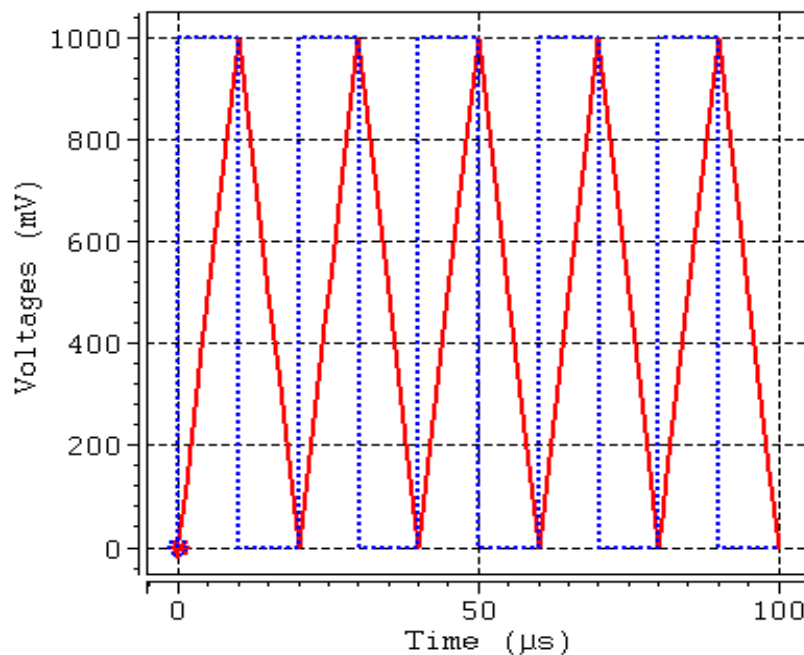


Fig. 11: Triangular to Square Wave Converter (waveforms shown on same axis).

CONCLUSION

Thus due to mapping of analog signals to their digital time domain provides the ease of modeling the mixed analog signals using Verilog-AMS which are more accurate and provide quick simulation results. A simple reliable, linear and highly symmetrical triangular-waveform, which can be operated in wide range of frequency, has been modeled in this article using the mixed signal modeling. This modeled behavior of generator exhibits many valuable properties, both with regard to the amplitude and frequency of the output waveforms. The ease, with which the frequency variation can be obtained, makes this modeled behavior to be more useful and user-friendly. The usability of modeling has been verified in different types of convertors which show appreciable results.

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